1. Clinical Question & Core Hypothesis

Clinical Question

Pretreatment cytolytic activity (CA)/high cytotoxic effector expression should predict higher rates of pCR to neoadjuvant chemotherapy in TNBC, but many CA-high/TIL-rich tumors nonetheless fail to achieve pCR — why is cytolytic transcriptional activation sometimes functionally ineffective?
2) Tier 1 paths used and why: I built on two complementary mechanistic axes: (a) cytotoxic effector execution (GZMB as a representative granzyme driving target-cell killing) and (b) IFN-γ signaling (IFNG) as a double-edged axis that both fosters immune recruitment/activation and induces adaptive resistance programs in tumor cells (e.g., checkpoint induction, antigen-presentation modulation). These two paths directly address the puzzle of ‘‘active-looking’’ immune transcriptional programs not yielding tumor eradication.
3) Tier 2 supporting evidence: CD8A, PRF1, GNLY, NKG7 and CTSW reinforce the cytotoxic module; PDCD1, CD274 (PD-L1), LAG3, TIGIT, STAT1, and CXCL9/10/11 describe IFN-driven recruitment and adaptive-exhaustion/resistance mechanisms; tumor-intrinsic modulators (HLA class I expression, antigen-presentation genes) mediate effectiveness of cytotoxicity. These support testing whether IFNG-associated adaptive programs blunt the cytolytic effect.
4) Why this endpoint: therapy_response (neoadjuvant chemotherapy pCR) is the most relevant clinical consequence for pretreatment CA in TNBC (well-supported in the literature, actionable for treatment decisions, and measurable in available NAC cohorts).
5) What I excluded and why: I deliberately excluded multi-gene poly-models and extra imaging features (TLS-like aggregates, TSR) to keep the mechanistic question sharp (one imaging axis + two molecular anchors). I also excluded metabolic/oxidative-stress modulators (e.g., NAMPT, SLC16A3) and broad proliferation markers from the core hypothesis to avoid diluting the causal claim linking cytolytic execution vs. IFN-driven adaptive resistance; those can be explored later as secondary modifiers if needed?

Working Hypothesis

In pretreatment TNBC patients undergoing neoadjuvant chemotherapy, high Tumor-Infiltrating Lymphocyte density with elevated GZMB denotes active cytotoxic engagement. However, concurrent high IFNG signaling induces adaptive resistance programs in tumor cells—including PD-L1 upregulation and alterations in antigen-presentation pathways—that blunt cytotoxic efficacy. As a result, tumors with high TIL density, high GZMB, and high IFNG activity will have lower therapy_response (pCR) rates to NAC than similarly profiled tumors with lower IFNG-driven adaptive resistance. This yields a testable interaction: IFNG status moderates the positive relationship between cytolytic activity (TIL density + GZMB) and therapy_response.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	GZMB, IFNG
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GZMB: GZMB as a representative granzyme driving target-cell killing) and (b) IFN-γ signaling (IFNG) as a double-edged axis that both fosters immune recruitment/activation and induces adaptive resistance programs in tumor cells (e.

IFNG: IFNG) as a double-edged axis that both fosters immune recruitment/activation and induces adaptive resistance programs in tumor cells (e.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CD8A	Supporting modifier from knowledge graph
Secondary	PRF1	Supporting modifier from knowledge graph
Secondary	GNLY	Supporting modifier from knowledge graph
Secondary	NKG7	Supporting modifier from knowledge graph
Secondary	CTSW	Supporting modifier from knowledge graph
Secondary	PDCD1	Supporting modifier from knowledge graph
Secondary	CD274	Supporting modifier from knowledge graph
Secondary	LAG3	Supporting modifier from knowledge graph
Secondary	TIGIT	Supporting modifier from knowledge graph
Secondary	STAT1	Supporting modifier from knowledge graph
Secondary	CXCL9	Supporting modifier from knowledge graph

Secondary	CXCL10	Supporting modifier from knowledge graph
Secondary	HLA_class_I	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CD8A: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PRF1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.